just mentioned—more precisely, as in Chapter 9, electrons and two kinds of quarks, up-quarks and down-quarks, that differ in mass and in electrical charge. But the experiments also revealed that the universe has other, more exotic particle species that don't arise in ordinary matter. In addition to up-quarks and down-quarks, experimenters have identified four other species of quarks (*charm-quarks, strange-quarks, bottom-quarks,* and *top-quarks*) and two other species of particles that are very much like electrons, only heavier (*muons* and *taus*). It is likely that these particles were plentiful just after the big bang, but today they are produced only as the ephemeral debris from high-energy collisions between the more familiar particle species. Finally, experimenters have also discovered three species of ghostly particles called *neutrinos* (*electron-neutrinos, muon-neutrinos,* and *tau-neutrinos*) that can pass through trillions of miles of lead as easily as we pass through air. These particles—the electron and its two heavier cousins, the six kinds of quarks, and the three kinds of neutrinos—constitute a modern-day particle physicist's answer to the ancient Greek question about the makeup of matter.[11]

The laundry list of particle species can be organized into three "families" or "generations" of particles, as in Table 12.1. Each family has two of the quarks, one of the neutrinos, and one of the electronlike particles; the only difference between corresponding particles in each family is that their masses increase in each successive family. The division into families certainly suggests an underlying pattern, but the barrage of particles can easily make your head spin (or, worse, make your eyes glaze over). Hang on, though, because one of the most beautiful features of string theory is that it provides a means for taming this apparent complexity.

According to string theory, there is only *one* fundamental ingredient—the string—and the wealth of particle species simply reflects the different vibrational patterns that a string can execute. It's just like what happens with more familiar strings like those on a violin or cello. A cello string can vibrate in many different ways, and we hear each pattern as a different musical note. In this way, one cello string can produce a range of different sounds. The strings in string theory behave similarly: they too can vibrate in different patterns. But instead of yielding different musical tones, *the different vibrational patterns in string theory*